Credit history

poor

Y	1
N	3
err	0.375

fair

Y	3
N	2
err	0.48

good

Y	5
N	1
err	0.28

$$\frac{0.375 \times 4 + 0.48 \times 5 + 0.28 \times 6}{15} = 0.372$$

age

Young

Y	2
N	4
err	0.44

Middle

Y	4
N	0
err	0

Senior

Y	3
N	2
err	0.48

$$\frac{0.44 \times 6 + 0.48 \times 5}{15} = 0.336$$

Income

High

Y	3
N	1
err	0.375

medium

Y	3
N	3
err	0.5

low

Y	3
N	2
err	0.48

$$\frac{0.375 \times 4 + 0.5 \times 6 + 0.48 \times 5}{15} = 0.46$$

own_home

yes

Y	7
N	1
ent	0.219

No

Y	2
N	5
ent	0.41

$$\frac{0.219 \times 8 + 0.41 \times 7}{15} = 0.31$$

Root node $\Rightarrow$ own_home.

own home $\Rightarrow$ yes!

age

Young

Y	2
N	1
ent	0.44

Middle

Y	3
N	0
ent	0

Senior

Y	2
N	0
ent	0

$$\frac{0.44 \times 3}{8} = 0.167$$

Credit history

poor

Y	1
N	1
ent	0.5

fair

Y	3
N	0
ent	0

good

Y	3
N	0
ent	0

$$\frac{0.5 \times 2}{8} = 0.125$$

Income

High

Y	3
N	0
err	0

pure

medium

Y	1
N	1
err	0.5

low

Y	3
N	0
err	0

pure.

$$\frac{0.5 \times 2}{8} = 0.125$$

Split on income!

Income → medium

age

Young

Y	0
N	1
err	0

Senior

Y	1
N	0
err	0

error = 0

Credit history

poor

Y	0
N	1
err	0

fair

Y	1
N	0
err	0

error = 0

Split on Credit Card history!

own_home ⇒ NO!

age

Young

Y	0
N	3
ent	0 pure

Middle

Y	1
N	0
ent	0 pure

Senior

Y	1
N	2
ent	0.444

$$\frac{0.44 \times 3}{7} = 0.19$$

Credit history

poor

Y	0
N	2
ent	0

fair

Y	0
N	2
ent	0

good

Y	2
N	1
ent	0.441

$$\frac{0.44 \times 3}{7} = 0.19$$

Income

High

Y	0
N	1
ent	0

medium

Y	2
N	2
ent	0.5

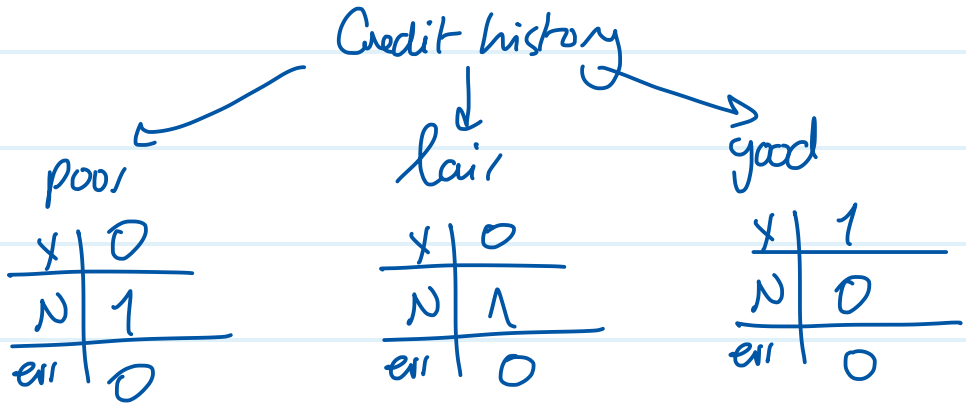
low

Y	0
N	2
ent	0

$$\frac{0.5 \times 4}{7} = 0.29$$

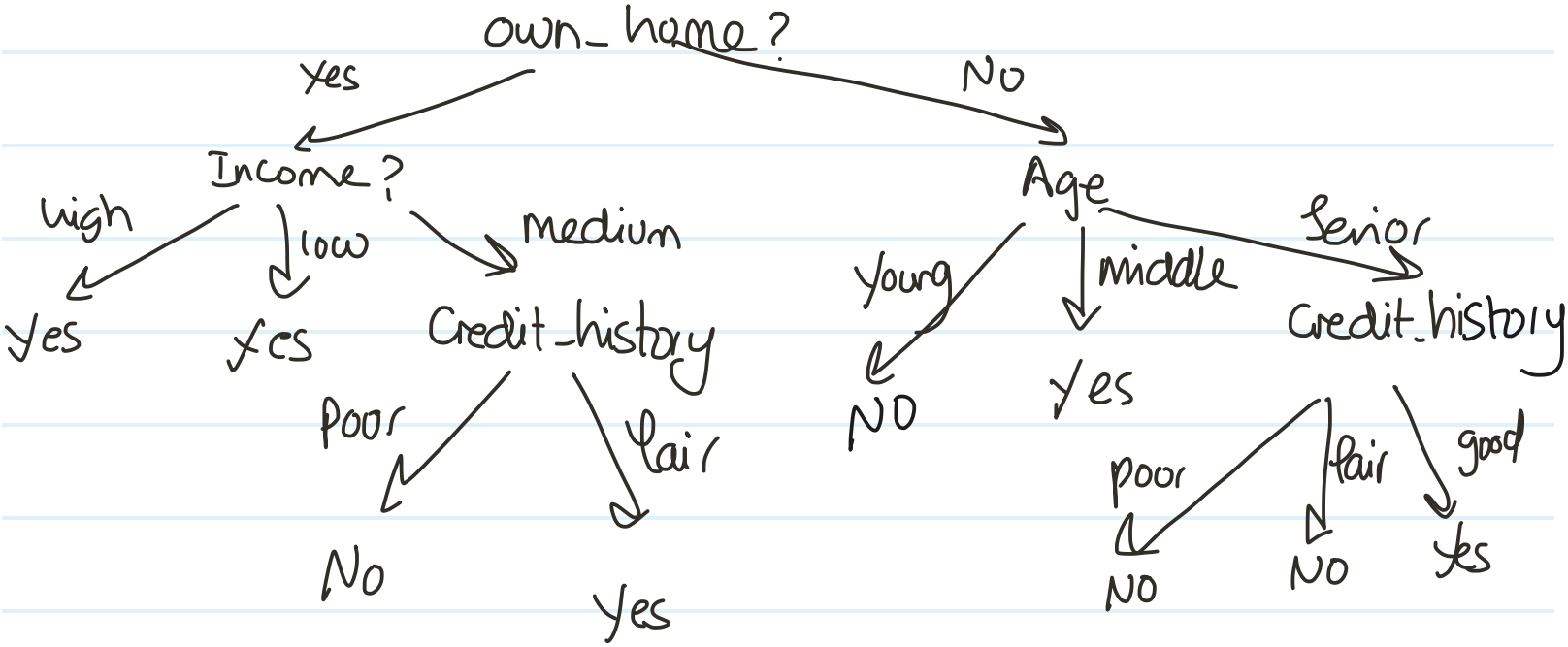
Split on age!!

age → Senior



error = 0

split on credit Card history.



A: NO

B: yes